PRACTICE SHEET

1. The value of $2^{2-\log_2 5}$ is equal to:
(a) 4/5 (b) 5/4
(c) 2/5 (d) 5/2

2. If $y = 2^{1/\log_x (8)}$, then x is equal to:
(a) y (b) y^2
(c) y^3 (d) None of these

3. If a, b and c are in GP, then $\log_{ax} x$, $\log_{bx} x$, $\log_{cx} x$ are in:
(a) GP (b) HP
(c) AP (d) None of these

4. If 2 log (x + 1) – log (x^2 – 1) = log2, then x equals to:
(a) 1 (b) 0
(c) 2 (d) 3

5. The number of solutions of $\log_2 (x-1) = 2\log_2 (x-3)$:
(a) 2 (b) 1
(c) 6 (d) 7

6. The value of $e^{(\log_{10} \tan 1^\circ + \log_{10} \tan 2^\circ + \ldots\ldots + \log_{10} \tan 89^\circ)}$ is equal to:
(a) 0 (b) 1
(c) e (d) 1/e

7. How many numbers of digits are there in 2^{98}? (Given that $\log_{10} 2 = 0.30103$)
(a) 98 (b) 99
(c) 30 (d) 29

8. If $\log_{30} 3 = a$ and $\log_{30} 5 = b$, what is the value of $\log_{30} 8$?
(a) 3(1 + a + b) (b) 3(1 – a + b)
(c) 3(1 + a – b) (d) 3(1 – a – b)

9. What is the least integral value of $2\log_{10} x - \log_x (0.01)$?
(a) 0 (b) 2
(c) 4 (d) 3

10. What is the value of $\log_{\sqrt{b}} a \log_{(c)^{1/3}} b \log_{(a)^{1/4}} c$?
(a) 12 (b) 24
(c) 1/12 (d) 1/24

11. If $\frac{\log x}{\log 5} = \frac{\log 36}{\log 6} = \frac{\log 64}{\log y}$, what are the values of x and y, respectively?
(a) 8, 25 (b) 25, 8
(c) 8, 8 (d) 25, 25

12. If $(\log_3 x)^2 + (\log_3 x) < 2$, then which one of the following is correct?
(a) $0 < x < \frac{1}{9}$ (b) $\frac{1}{9} < x < 3$
(c) $3 < x < \infty$ (d) $\frac{1}{9} \le x \le 3$

13. The value of $7 \log \frac{16}{15} + 5 \log \frac{25}{24} + 3 \log \frac{81}{80}$ is :
(a) 0.3010 (b) 0.3512
(c) 0.412 (d) None of the above

14. The value of $\log \frac{70}{33} + \log \frac{22}{135} - \log \frac{7}{18}$ is:
(Given log 2 = 0.3010, log 3 = 0.4771)
(a) – 0.0512 (b) 0.4213
(c) 0.3010 (d) None of these

15. The value of $3^{2-\log_3 (4\log 36)}$ is:
(Given log 2 = 0.3010, log 3 = 0.4771)
(a) 1.45 (b) 0.69
(c) 1.2 (d) 3.2

16. If $\log_{10} \sqrt{1+x} + 3\log_{10} \sqrt{1-x} = \log_{10} \sqrt{1-x^2} + 2$, then the value of x is:
(a) 4 (b) –99
(c) 5 (d) None of these

17. $\log_{25} 625 - \log_{31} 961 + \log_{29} 841 =$
(a) 2 (b) 6
(c) 0 (d) 4

18. If log x + log y = log (x + y) then :
(a) x = y (b) xy = 1
(c) $y = \frac{x-1}{x}$ (d) $y = \frac{x}{x-1}$

19. If a , b , c are in GP then $\log_a n$, $\log_b n$, $\log_c n$ are in
(a) A.P. (b) G.P.
(c) H.P. (d) not

20. x triples every second . How will $\log_2 x$ change every second
(a) It will double every second
(b) It will triple every second
(c) It increases by constant amount every second
(d) None of these

21. If $\log_3 \log_3 [\log_3 x] = \log_3 3$, then what is the value of x?
(a) 3 (b) 27
(c) 3^9 (d) 3^{27}

22. What is the value of $\frac{\log_{\sqrt{\alpha\beta}} H}{\log_{\sqrt{\alpha\beta\gamma}} H}$?
(a) $\log_{\alpha\beta} (\alpha)$ (b) $\log_{\alpha\beta\gamma} (\alpha\beta)$
(c) $\log_{\alpha\beta} (\alpha\beta\gamma)$ (d) $\log_{\alpha\beta} (\beta)$

23. The value of x for which
$\log_9 x - \log_9 \left(\frac{x}{10} + \frac{1}{9}\right) = 1$ is:
(a) 2 (b) 4
(c) 9 (d) 10

24. If $\log_4(x^2 + x) - \log_4(x + 1) = 2$, then the value of x is:
(a) 2 (b) 4
(c) 5 (d) 16

25. The value of $\left(\frac{1}{\log_3 60} + \frac{1}{\log_4 60} + \frac{1}{\log_5 60}\right)$ is:
(a) 60 (b) 5
(c) 1 (d) 0

26. The value of:

$\left[\frac{1}{\log_{(p/q)} x} + \frac{1}{\log_{(q/r)} x} + \frac{1}{\log_{(r/p)} x}\right]$ is :

(a) 3 (b) 2
(c) 1 (d) 0

27. If $\log_{10} 2 = 0.3010$ and $\log_{10} 3 = 0.4771$, then the value of $\log_{10}(1.5)$ is:
(a) 0.7161 (b) 0.1761
(c) 0.7116 (d) 0.7611

28. Given log 2 = 0.3010, the number of digits in 5^{20} is:
(a) 14 (b) 16
(c) 18 (d) 25

29. If x = 27 and y = $\log_3 4$, then x^y = ?
(a) 1/16 (b) 3/7
(c) 16 (d) 64

30. If $x^{18} = y^{21} = z^{28}$, then 3, $3\log_y x$, $3\log_z y$, $7\log_x z$ are in:
(a) A.P. (b) G.P.
(c) H.P. (d) none of these

31. If $\log_3 x + \log_9 x^2 + \log_{27} x^3 = 3$, then the value of x is:
(a) 6 (b) 3
(c) 1 (d) 2

32. If $\log_{10}(x - 15) + \log_{10} 10 = 4$, then the value of x is:
(a) 785 (b) 985
(c) 1015 (d) None

ANSWER KEY

1.	a	**2.**	c	**3.**	b	**4.**	d	**5.**	b	**6.**	b	**7.**	c	**8.**	d	**9.**	c	**10.**	b
11.	b	**12.**	b	**13.**	a	**14.**	a	**15.**	a	**16.**	d	**17.**	a	**18.**	d	**19.**	c	**20.**	c
21.	d	**22.**	c	**23.**	d	**24.**	d	**25.**	c	**26.**	d	**27.**	b	**28.**	a	**29.**	d	**30.**	a
31.	b	**32.**	c																

Solutions

Sol. 1. (a)

$2^{2-\log_2 5} = \frac{2^2}{2^{\log_2 5}} = \frac{4}{5}$

Sol. 2. (c)
taking log both side

$y = 2^{\frac{1}{\log_x 8}} \Rightarrow \log_x y = \frac{1}{\log_x 8} \times \log_x 2$

$\Rightarrow \log_x y = \log_8 2 = 1/3$
$\Rightarrow x = y^3$

Sol. 3. (b)
a,b,c are in G.P. $\Rightarrow b^2 = ac$
taking log both sides 2log b = log a + log c
i.e. $\log_x a$, $\log_x b$, $\log_x c$ are in AP
$\Rightarrow 1 + \log_x a$, $1 + \log_x b$, $1 + \log_x c$ are in AP
$\Rightarrow \log_x x + \log_x a$, $\log_x x + \log_x b$, $\log_x x + \log_x c$ are in AP
$\Rightarrow \log_x ax$, $\log_x bx$, $\log_x cx$ are in AP
$\Rightarrow \log_{ax} x$, $\log_{bx} x$, $\log_{cx} x$ are in HP

Sol. 4. (d)
$2\log(x+1) - \log(x^2-1) = \log 2$

$\text{Log}\frac{(x+1)^2}{x^2-1} = \log 2 \Rightarrow \frac{(x+1)^2}{x^2-1} = 2$

$\Rightarrow \frac{x+1}{x-1} = 2 \Rightarrow x = 3$

Sol. 5. (b)
$\log_2(x-1) = 2\log_2(x-3)$
$\log_2(x-1) = \log_2(x-3)^2$
$\Rightarrow x - 1 = (x-3)^2 \Rightarrow x - 1 = x^2 - 6x + 9$
$\Rightarrow x^2 - 7x + 10 = 0 \Rightarrow x = 2, 5$
but x = 2 is not in domain so there is only one solution.

Sol. 6. (b)

$e^{(\log_{10}\tan 1° + \log_{10}\tan 2° + ... + \log_{10}\tan 89°)}$

$\Rightarrow e^{\log_{10}\tan 1°.\tan 2°.\tan 3°........\tan 89°}$

$= e^{\log_{10} 1} = e^0 = 1$

Sol. 7. (c)
No. of digits = 98 $\log_{10} 2$
= 98 × 0.3010 = 29.49 ≈30 digits

Sol. 8. (d)

$\log_{30} 8 = 3\log_{30} 2 = 3\log_{30}\frac{30}{15}$

$\Rightarrow 3\log_{30}\frac{30}{3\times 5} \Rightarrow 3(\log_{30} 30 - \log_{30} 3 - \log_{30} 5)$

$\Rightarrow 3(1 - a - b)$

Sol. 9. (c)
$2\log_{10} x - \log_x (0.01)$
$\Rightarrow 2\log_{10} x - \log_x (10)^{-2}$
$\Rightarrow 2\log_{10} x + 2\log_x 10$
$\Rightarrow 2(\log_{10} x + \log_x 10)$
We know that

$\Rightarrow x + \frac{1}{x} \geq 2$ [By the concept of AM ≥ GM]

So $\log_{10} x + \log_x 10 \geq 2$
$\Rightarrow 2\log_{10} x - \log_x 0.01 \geq 4$

Sol. 10. (b)

$= \log_{\sqrt{b}} a . \log_{\sqrt[3]{c}} b . \log_{\sqrt[4]{a}} c$

$= 2\log_b a . 3\log_c b . 4\log_a c = 24$

Sol. 11. (b)

$\frac{\log x}{\log 5} = \frac{\log 36}{\log 6} = \frac{\log 64}{\log y}$,

$\Rightarrow \log_5 x = \log_6 36 = \log_y 64$
$\Rightarrow \log_5 x = 2 = \log_y 64$ x = 25, y = 8

Sol. 12. (b)
$\Rightarrow (\log_3 x)^2 + (\log_3 x) < 2$
$\Rightarrow (\log_3 x)^2 + (\log_3 x) - 2 < 0$
$\Rightarrow (\log_3 x + 2)(\log_3 x - 1) < 0$
$\Rightarrow -2 < \log_3 x < 1$
$\Rightarrow 1/9 < x < 3$

Sol. 13. (a)

$\Rightarrow 7\log\frac{16}{15} + 5\log\frac{25}{24} + 3\log\frac{81}{80}$

$\log\left(\frac{2^4}{3\times 5}\right)^7 + \log\left(\frac{5^2}{2^3\times 3}\right)^5 + \log\left(\frac{3^4}{2^4\times 5}\right)^3$

$\Rightarrow \log\left(\frac{2^{28}}{3^7\times 5^7}.\frac{5^{10}}{2^{15}\times 3^5}.\frac{3^{12}}{2^{12}\times 5^3}\right)$

$\Rightarrow \log 2 = 0.3010$

Sol. 14. (a)

$\log\frac{70}{33} + \log\frac{22}{135} - \log\frac{7}{18}$

$\Rightarrow \log\left(\frac{\frac{70}{33}.\frac{22}{135}}{\frac{7}{18}}\right) = \log\frac{8}{9} = \log 8 - \log 9$

$\Rightarrow 3\log 2 - 2\log 3 = 3(0.3010) - 2(0.4771)$
$= -0.0512$

Sol. 15. (a)

$3^{2-\log_3(4\log 36)} = \frac{3^2}{3^{\log_3(4\log 36)}} = \frac{9}{4\log 36}$

$= \frac{9}{8\log 6} = \frac{9}{8(\log 2 + \log 3)} = \frac{9}{8\times 0.7781} = 1.45$

Sol. 16. (d)

$\log_{10}\sqrt{1+x} + 3\log_{10}\sqrt{1-x}$

$= \log_{10}\sqrt{1-x^2} + 2$

$\log\left(\sqrt{x+1}\right)\left(\sqrt{1-x}\right)^3 = \log 100\sqrt{1-x^2}$

$\Rightarrow \sqrt{1-x^2}(1-x) = 100\sqrt{1-x^2}$

$\Rightarrow \sqrt{1-x^2}(1-x) - 100\sqrt{1-x^2} = 0$

$= \sqrt{1-x^2}(x+99) \Rightarrow x = 1, -1 - 99$
but all these values are not in domain.

Sol. 17. (a)
$\log_{25} 625 - \log_{31} 961 + \log_{29} 841$
$= 2 - 2 + 2 = 2$

Sol. 18. (d)
log x + log y = log (x + y)
⇒ log(xy) = log (x + y)
⇒ xy = x + y
$y = \frac{x}{x-1}$

Sol. 19. (c)
a, b, c are in GP
then $b^2 = ac$
taking log both sides
$2\log_n b = \log_n a + \log_n c$
so $\log_n a$, $\log_n b$, $\log_n c$ are in AP then
$\log_a n$, $\log_b n$, $\log_c n$ are in H.P.

Sol. 20. (c)
x become 3x then
logx become log3x or log3 + logx
so it will increase by a constant value log3

Sol. 21. (d)
$\log_3 \log_3[\log_3 x] = \log_3 3$
⇒ $\log_3 [\log_3 x] = 3$
⇒ $[\log_3 x] = 27$
$x = 3^{27}$

Sol. 22. (c)
$$\frac{\log_{\sqrt{\alpha\beta}} H}{\log_{\sqrt{\alpha\beta\gamma}} H} = \frac{\frac{\log H}{\log\sqrt{\alpha\beta}}}{\frac{\log H}{\log\sqrt{\alpha\beta\gamma}}} = \frac{\log\sqrt{\alpha\beta\gamma}}{\log\sqrt{\alpha\beta}}$$
$$\Rightarrow \frac{\frac{1}{2}\log(\alpha\beta\gamma)}{\frac{1}{2}\log(\alpha\beta)} = \frac{\log(\alpha\beta\gamma)}{\log(\alpha\beta\gamma)} = \log_{(\alpha\beta)}(\alpha\beta\gamma)$$

Sol. 23. (d)
Given equation may be written as:
$$\log_9 x - \log_9\left(\frac{x}{10}+\frac{1}{9}\right) = \log_9 9$$
$$\Rightarrow \log_9 \frac{x}{\left(\frac{x}{10}+\frac{1}{9}\right)} = \log_9 9$$
$$\Rightarrow \frac{x}{\left(\frac{x}{10}+\frac{1}{9}\right)} = 9$$
$$\Rightarrow x = \frac{9x}{10} + 1$$
⇒ x = 10

Sol. 24. (d)
$\log_4 (x^2 + x) - \log_4 (x + 1) = 2 \log_4 4$
$$\Rightarrow \log_4\left(\frac{x^2+x}{x+1}\right) = \log_4 (4^2)$$
$$\Rightarrow \frac{x(x+1)}{(x+1)} = 16$$
⇒ x = 16

Sol. 25. (c)
Given Exp.
⇒ $\log_{60} 3 + \log_{60} 4 + \log_{60} 5$
⇒ $\log_{60} (3 \times 4 \times 5) = \log_{60} 60 = 1$

Sol. 26. (d)
Given Exp.
$$\log_x\left(\frac{p}{q}\right) + \log_x\left(\frac{q}{r}\right) + \log_x\left(\frac{r}{p}\right)$$
$$= \log_x\left(\frac{p}{q}\times\frac{q}{r}\times\frac{r}{p}\right)$$
$= \log_x 1 = 0$

Sol. 27. (b)
$\log_{10} (1.5) = \log_{10} (3/2)$
$= \log_{10} 3 - \log_{10} 2$
= (0.4771 – 0.3010) = 0.1761

Sol. 28. (a)
$\log 5^{20}$ 20 = 20log 5
= 20log (10/2)
= 20 × [log 10 – log 2]
= 20 × (1 – 0.3010)
= 20 × 0.6990 = 13.98
Characteristic = 14
Number of digits = 14

Sol. 29. (d)
$x^y = (27)^{\log_3 4} = (4)^{\log_3 27} = (4)^3 = 64$

Sol. 30. (a)
Let $x^{18} = y^{21} = z^{28}$ = k. then
18 log x = 21 log y = 28 log z = log k
$\log x = \frac{\log k}{18}, \log y = \frac{\log k}{21}, \log z = \frac{\log k}{28}$
$$\Rightarrow 3\log_y x = \frac{3\log x}{\log y} = \frac{3\times 21}{18} = \frac{7}{2},$$
$$\Rightarrow 3\log_z y = \frac{3\log y}{\log z} = \frac{3\times 28}{28} = 4$$
$$\Rightarrow 7\log_x z = \frac{7\log z}{\log x} = \frac{7\times 28}{28} = \frac{9}{2}$$
3,7/2, 4,9/2 are in A.P.

Sol. 31. (b)
Using $\log_{(a^n)} x = \frac{1}{n}\log_a x$, we have
$$\log_{3^2} x^2 = \frac{1}{2}\log_3 x^2 = \log_3 x$$
$$\log_{3^3} x^3 = \frac{1}{3}\log_3 x^3 = \log_3 x$$
Given equ. is
$3\log_3 x = 3$
⇒ $\log_3 x = 1 = \log_3 3$
⇒ x = 3

Sol. 32. (c)
$\log_{10} (x - 15) + \log_{10} 10 = 4$
⇒ $\log_{10} (x - 15) + 1 = 4$
⇒ $\log_{10} (x - 15) = 3$
⇒ $(x - 15) = 10^3 = 1000$
⇒ x = 1015

OFFICIUM SUPREMUM

NDA PYQ

1. What is $\log\left(a+\sqrt{a^2+1}\right)+\log\left(\frac{1}{a+\sqrt{a^2+1}}\right)$ equal to?

(a) 1 (b) 0

(c) 2 (d) $\frac{1}{2}$

[NDA (I) 2011]

2. What is the value of:

$\log_{10}\left(\frac{9}{8}\right)-\log_{10}\left(\frac{27}{32}\right)+\log_{10}\left(\frac{3}{4}\right)$

(a) 3 (b) 2

(c) 1 (d) 0

[NDA (I) 2011]

3. What is the value of $\log_2(\log_3 81)$?

(a) 2 (b) 3

(c) 4 (d) 9

[NDA (II) 2011]

4. If $\log_{10} 2$, $\log_{10}(2^x-1)$, $\log_{10}(2^x+3)$ are three consecutive terms of an AP, then which one of the following is correct?

(a) $x = 0$ (b) $x = 1$

(c) $x = \log_2 5$ (d) $x = \log_5 2$

[NDA (II)-2011]

5. What is the value of $2\log_8 2-\frac{1}{3}\log_3 9$

(a) 0 (b) 1

(c) 2 (d) 1/3

[NDA (I) 2012]

6. If $(\log_3 x)(\log_x 2x)(\log_{2x} y) = \log_x x^2$, then what is the value of y?

(a) 4.5 (b) 9

(c) 18 (d) 27

[NDA (II) 2012]

7. What is the value of $\log_y x^5 \log_x y^2 \log_z z^3$?

(a) 10 (b) 20

(c) 30 (d) 60

[NDA (I) 2013]

8. What is $\log_{81} 243$ equal to?

(a) 0.75 (b) 1.25

(c) 1.5 (d) 3

[NDA (II) 2013]

9. If $\log_{10} 2$, $\log_{10}(2^x - 1)$ and $\log_{10}(2^x + 3)$ are three consecutive terms of an AP, then the value of x is:

(a) 1 (b) $\log_5 2$

(c) $\log_2 5$ (d) $\log_{10} 5$

[NDA (I) 2015]

10. If $\log_8 m + \log_8 \frac{1}{6}=\frac{2}{3}$ then m is equal to?

(a) 24 (b) 18

(c) 12 (d) 4

[NDA (II) 2015]

11. If $\log_a(ab) = x$, then what is $\log_b(ab)$ equal to?

(a) $\frac{1}{x}$ (b) $\frac{x}{x+1}$

(c) $\frac{x}{1-x}$ (d) $\frac{x}{x-1}$

[NDA (I) 2016]

12. If $x + \log_{10}(1+2^x) = x\log_{10}5+\log_{10}6$ then x is equal to

(a) 2, –3 (b) 2 only

(c) 1 (d) 3

[NDA (II) 2017]

13. The value of $\frac{1}{\log_3 e}+\frac{1}{\log_3 e^2}+\frac{1}{\log_3 e^4}+\ldots$..up to infinite terms:

(a) $\log_e 9$ (b) 0

(c) 1 (d) loge 3

[NDA (II) 2017]

14. If $x + \log_{15}(1 + 3^x) = x\log_{15}5+\log_{15}12$, where x is an integer, then what is x equal to?

(a) –3 (b) 2

(c) 1 (d) 3

[NDA (I) 2018]

15. If n = (2017)! then what is

$\frac{1}{\log_2 n}+\frac{1}{\log_3 n}+\frac{1}{\log_4 n}+\ldots\ldots\frac{1}{\log_{2017} n}$ equal to?

(a) 0 (b) 1

(c) n/2 (d) n

[NDA (I) 2018]

16. What is $\frac{1}{\log_2 N}+\frac{1}{\log_3 N}+\frac{1}{\log_4 N}+\ldots\ldots\frac{1}{\log_{100} N}$ equal to:

(a) $\frac{1}{\log_{100!} N}$ (b) $\frac{1}{\log_{99!} N}$

(c) $\frac{99}{\log_{100!} N}$ (d) $\frac{99}{\log_{99!} N}$

[NDA (I) 2018]

17. If $0 < a < 1$, the value of $\log_{10} a$ is negative. This is justified by:

(a) Negative power of 10 is less than 1

(b) Negative power of 10 is between 0 and 1

(c) Negative power of 10 is positive

(d) Negative power of 10 is negative

[NDA-2018(1)]

18. $\text{Log}_7 \text{Log}_7 \sqrt{7\sqrt{7\sqrt{7}}}$?

(a) $3\log_2 7$ (b) $1 - 3\log_2 7$

(c) $1-3\log_7 2$ (d) 7/8

[NDA (II) 2018]

19. What is the value of $\log_9 27 + \log_8 32$?

(a) 7/2 (b) 19/6

(c) 4 (d) 7

[NDA (II) 2018]

20. If $(0.2)^x = 2$ and $\log_{10} 2 = 0.3010$, then what is the values of x nearest tenth?

(a) –10 (b) –0.5

(c) –0.4 (d) – 0.2

[NDA (II) 2018]

21. If $f(x) = \log_{10}(1 + x)$,than what is $4f(4) + 5f(1) - \log_{10}2$ equal to ?

(a) 0 (b) 1

(c) 2 (d) 4

[NDA (I) 2019]

22. $f(x) = \log_x 10$ is defined in the domain

(a) $x > 10$ (b) $x > 0$, except $x = 10$

(c) $x \geq 10$ (d) $x > 0$, except $x = 1$

[NDA (I) 2019]

23. If $x^{\log_7 x} > 7$ where x > 0, then which one of the following is correct?
(a) $x \in (0,\infty)$ (b) $x \in (1/7,7)$
(c) $x \in (0,1/7) \cup (7,\infty)$ (d) $x \in (1/7,\infty)$

[NDA (II) 2019]

24. What is the value of
$\frac{1}{10}\log_5 1024 - \log_5 10 + \frac{1}{5}\log_5 3125$?
(a) 0 (b) 1
(c) 2 (d) 3

[NDA 2020]

25. If $x = \log_c(ab), y = \log_a(bc), z = \log_b(ac)$, then which one of the following is correct?
(a) xyz = 1
(b) x + y + z = 1
(c) $(1 + x)^{-1} + (1 + y)^{-1} + (1 + z)^{-1} = 1$
(d) $(1 + x)^{-2} + (1 + y)^{-2} + (1 + z)^{-2} = 1$

[NDA 2020]

26. The value of x , satisfying the equation $\log_{\cos x} \sin x = 1$, where $0 < x < \frac{\pi}{2}$, is:
(a) $\frac{\pi}{12}$ (b) $\frac{\pi}{3}$
(c) $\frac{\pi}{4}$ (d) $\frac{\pi}{6}$

[NDA (I) 2021]

27. If n = (100)! then what is
$\frac{1}{\log_2 n} + \frac{1}{\log_3 n} + \frac{1}{\log_4 n} + \dots\dots + \frac{1}{\log_{100} n}$ equal to?
(a) 0 (b) 1
(c) 2 (d) 3

[NDA (I) 2021]

28. If $\log_{10} 2 \log_2 10 + \log_{10} (10^x) = 2$, then what is the value of x?
(a) 0 (b) 1
(c) $\log_2 10$ (d) $\log_5 2$

[NDA (II) 2021]

29. If $\log_{10}2$, $\log_{10}(2^x - 1)$, $\log_{10}(2^x + 3)$ are in AP, then what is x equal to?
(a) 0 (b) 1
(c) $\log_2 5$ (d) $\log_5 2$

[NDA (II) 2021]

30. If $\log_x a$, a^x and $\log_b x$ are in GP, then what is x equal to?
(a) $\log_a (\log_a b)$ (b) $\log_b (\log_a b)$
(c) $\frac{\log_a(\log_b a)}{2}$ (d) $\frac{\log_b(\log_a b)}{2}$

[NDA – 2023 (1)]

31. What is the number of solutions of $\log_4(x - 1) = \log_2(x - 3)$?
(a) Zero (b) One
(c) Two (d) Three

[NDA – 2024 (1)]

32. For $x \geq y > 1$,
Let $\log_x\left(\frac{x}{y}\right) + \log_y\left(\frac{y}{x}\right) = k$,
Then the value of k can never be equal to
(a) -1 (b) $-\frac{1}{2}$
(c) 0 (d) 1

[NDA – 2024 (1)]

33. If $\log_b a = p$, $\log_d c = 2p$ and $\log_f e = 3p$, then what is $(ace)^{\frac{1}{p}}$ equal to?
(a) bd^2f^3 (b) bdf
(c) b^3d^3f (d) $b^2d^2f^2$

[NDA – 2024 (1)]

Direction for next two:

Let $p = \sum_{j=1}^{n} \log_{10} 2^j$ and $q = \sum_{j=1}^{n} \log_{10} 5^j$

[NDA – 2025 (2)]

34. If p + q = 66, then which one of the following is correct?
(a) n < 7 (b) 7 < n < 9
(c) 9 < n < 12 (d) n > 12

35. If p + q = 15, then what is q – p equal to?
(a) $\log_{10} 2.5$ (b) $5\log_{10} 2.5$
(c) $10\log_{10} 2.5$ (d) $15\log_{10} 2.5$

ANSWER KEY

1.	b	**2.**	d	**3.**	a	**4.**	c	**5.**	a	**6.**	b	**7.**	c	**8.**	b	**9.**	c	**10.**	a
11.	d	**12.**	c	**13.**	a	**14.**	c	**15.**	b	**16.**	a	**17.**	b	**18.**	c	**19.**	b	**20.**	c
21.	d	**22**	d	**23.**	c	**24.**	a	**25.**	c	**26.**	c	**27.**	b	**28.**	b	**29.**	c	**30.**	c
31.	b	**32.**	d	**33.**	a	**34.**	c	**35.**	d										

Solutions

Sol.1. (b)

$$\log\left(a+\sqrt{a^2+1}\right)+\log\left(\frac{1}{a+\sqrt{a^2+1}}\right)$$

$$\log\left(a+\sqrt{a^2+1}\right)\left(\frac{1}{a+\sqrt{a^2+1}}\right)=\log 1=0$$

Sol.2. (d)

$$\log_{10}\left(\frac{9}{8}\right)-\log_{10}\left(\frac{27}{32}\right)+\log_{10}\left(\frac{3}{4}\right)$$

$$\log_{10}\left(\frac{\frac{9}{8}\cdot\frac{3}{4}}{\frac{27}{32}}\right)=\log_{10}1=0$$

Sol.3. (a)

$\log_2(\log_3 81)=\log_2 4=2$

Sol.4. (c)

$\log_{10} 2$, $\log_{10}(2^x-1)$, $\log_2)2^x+3)$ are in AP

$2\log_{10}(2^x-1)=\log_{10}2+\log_{10}(2^x+3)$

$\Rightarrow (2^x-1)^2=2.(2^x+3)$

$\Rightarrow (2^x)^2-2.2^x+1=2.2^x+6$

$\Rightarrow (2^x)^2-4.2^x-5=0$

$\Rightarrow (2^x-5)(2^x+1)=0$

$2^x=5$ and $2^x=-1$

$2^x=-1$ is Not possible

$x=\log_2 5$

Sol.5. (a)

$$2\log_8 2-\frac{1}{3}\log_3 9=\frac{2}{3}\log_2 2-\frac{2}{3}\log_3 3=0$$

Sol.6. (b)

$\log_3 x)(\log_x 2x)(\log_{2x} y)=\log_x x$

$$\frac{\log x}{\log 3}\frac{\log 2x}{\log x}\frac{\log y}{\log 2x}=2$$

$$\frac{\log y}{\log 3}=2\Rightarrow \log_3 y=2\Rightarrow y=3^2=9$$

Sol.7. (c)

$\log_y x^5 \log_x y^2 \log_z z^3$

$\Rightarrow 5\log_y x.2\log_x y.3\log_z z$

$\Rightarrow 30\log_y x.\log_x y.\log_z z=30$

Sol.8. (b)

$$\log_{81}243=\log_{3^4}3^5=\frac{5}{4}\log_3 3=\frac{5}{4}=1.25$$

Sol.9. (c)

$2\log_{10}(2^x-1)=\log_{10}2+\log_{10}(2^x+3)$

$2\log_{10}(2^x-1)=\log_{10}2.(2^x+3)$

$\Rightarrow (2^x-1)^2=2.(2^x+3)$

$\Rightarrow 2^{2x}-2.2^x+1=2.2^x+6$

$\Rightarrow 2^{2x}-4.2^x-5=0$

$\Rightarrow (2^x-5)(2^x+1)=0$

$2^x=5\Rightarrow x=\log_2 5$

Sol.10. (a)

$\mathrm{Log}_8 m+\log_8\frac{1}{6}=\frac{2}{3}$

$\log_8\frac{m}{6}=\frac{2}{3}\Rightarrow\frac{m}{6}=8^{2/3}=4$

$m=24$

Sol.11. (d)

$\log_a(ab)=x\Rightarrow\log_a a+\log_a b=x$

$\Rightarrow 1+\log_a b=x\Rightarrow\log_a b=x-1$

$\Rightarrow \log_b a=\frac{1}{x-1}$

$\Rightarrow \log_b a+\log_b b=\frac{1}{x-1}+1$

$\Rightarrow \log_b(ab)=\frac{x}{x-1}$

Sol.12. (c)

$\Rightarrow x+\log_{10}(1+2^x)=x\log_{10}5+\log_{10}6$

$\Rightarrow x\log_{10}10+\log_{10}(1+2^x)=x\log_{10}5+\log_{10}6$

$\Rightarrow \log_{10}10^x+\log_{10}(1+2^x)=\log_{10}5^x+\log_{10}6$

$\Rightarrow \log_{10}10^x.(1+2^x)=\log_{10}5^x.6$

$\Rightarrow 10^x.(1+2^x)=5^x.6$

$\Rightarrow 2^x.(1+2^x)=6$

$\Rightarrow 2^{2x}+2^x-6=0$

$\Rightarrow (2^x+3)(2^x-2)=0$

$\Rightarrow x=1$

Sol.13. (a)

$$\frac{1}{\log_3 e}+\frac{1}{\log_3 e^2}+\frac{1}{\log_3 e^4}+......$$

$$=\log_e 3+\log_{e^2}3+\log_{e^4}3+......$$

$$=\left(1+\frac{1}{2}+\frac{1}{4}+.......\right)\log_e 3$$

$=2\log_e 3=\log_e 9$

Sol.14. (c)

$x+\log_{15}(1+3^x)=x\log_{15}5+\log_{15}12$

$x\log_{15}15+\log_{15}(1+3^x)=x\log_{15}5+\log_{15}12$

$\log_{15}15^x+\log_{15}(1+3^x)=\log_{15}5^x+\log_{15}12$

$\log_{15}[15^x(1+3^x)]=\log_{15}5^x.12$

$3^x.5^x(1+3^x)=5^x.12$

$3^{2x}+3^x-12=0$

Put $3^x=y$

$y^2+y-12=0$

$y=3\Rightarrow 3^x=3\Rightarrow x=1$

$y=-4\Rightarrow 3^x=-4$ (not possible)

Sol.15. (b)

$$\frac{1}{\log_2 n}+\frac{1}{\log_3 n}+\frac{1}{\log_4 n}+.......+\frac{1}{\log_{2017}n}$$

$\log_n 2+\log_n 3+\log_n 4+\ldots\ldots.\log_n 2017$

$\log_n 2.3.4\ldots\ldots..2017=\log_n(2017)!$

$=\log_{(2017)!}(2017)!=1$

Sol.16. (a)

$$\frac{1}{\log_2 N}+\frac{1}{\log_3 N}+\frac{1}{\log_4 N}+......+\frac{1}{\log_{100}N}$$

$\log_N 2+\log_N 3+\log_N 4+\ldots\ldots\log_N 100$

$\log_N 2.3.4\ldots\ldots..100=\log_N(100)!$

$$=\frac{1}{\log_{(100)!}N}$$

Sol.17. (b)

Negative power of 10 is lie between 0 and 1

Sol.18. (c)

$\log_7\log_7 7^{7/8}$

$$\log_7\left(\frac{7}{8}\log_7 7\right)=\log_7\frac{7}{8}=\log_7 7-\log_7 8$$

$\Rightarrow 1-\log_7 8=1-3\log_7 2$

Sol.19. (b)

$\log_9 27+\log_8 32$

$$=\log_{3^2}3^3+\log_{2^3}2^5=\frac{3}{2}+\frac{5}{3}=\frac{19}{6}$$

Sol.20. (c)

$(0.2)^x=2$

taking log both sides

$\Rightarrow x\log 0.2=\log 2$

$$x=\frac{\log 2}{\log 0.2}=\frac{\log 2}{\log 2/10}=\frac{\log 2}{\log 2-\log 10}$$

$$=\frac{0.3010}{0.3010-1}=\frac{0.3010}{-0.6990}\approx-0.4$$

Sol.21. (d)

$f(x)=\log_{10}(1+x)$

$4f(4)+5f(1)-\log_{10}2$

$\Rightarrow 4\log_{10}5+5\log_{10}2-\log_{10}2$

$\Rightarrow 4\log_{10}5+4\log_{10}2=4\log_{10}10=4$

Sol.22. (d)

Base of log cannot be 1

Sol.23. (c)

$x^{\log_7 x}>7$

taking log both sides

$\Rightarrow \log_7 x^{\log_7 x}>\log_7 7$

$\Rightarrow (\log_7 x)^2>1$

$\Rightarrow \log_7 x<-1$ or $\log_7 x>1$

$\Rightarrow x<-1/7$ or $x>7$

Sol.24. (a)

$$\frac{1}{10}\log_5 1024-\log_5 10+\frac{1}{5}\log_5 3125$$

$$=\frac{1}{10}\log_5 2^{10}-\log_5 10+\frac{1}{5}\log_5 5^5$$

$$=\frac{10}{10}\log_5 2-\log_5 10+\frac{5}{5}\log_5 5$$

$=\log_5 2-\log_5 10+\log_5 5=\log_5\frac{2\times 5}{10}=0$

Sol.25. (c)

$$\frac{1}{1+x}+\frac{1}{1+y}+\frac{1}{1+z}$$

$$=\frac{1}{1+\log_c ab}+\frac{1}{1+\log_a bc}+\frac{1}{1+\log_b ac}$$

$$\frac{1}{\log_c c+\log_c ab}+\frac{1}{\log_a a+\log_a bc}+\frac{1}{\log_b b+\log_b ac}$$

$$\Rightarrow\frac{1}{\log_c abc}+\frac{1}{\log_a abc}+\frac{1}{\log_b abc}$$

$\Rightarrow \log_{abc}c+\log_{abc}b+\log_{abc}a$

$\Rightarrow \log_{abc}abc=1$

Sol.26. (c)

$\log_{\cos x}\sin x=1$

$\frac{\log\sin x}{\log\cos x}=1\Rightarrow\log\sin x=\log\cos x$

$\Rightarrow \log\sin x-\log\cos x=0$

$\Rightarrow \log\frac{\sin x}{\cos x}=\log\tan x=0$

$\Rightarrow x=\frac{\pi}{4}$

Sol.27. (b)

$$\frac{1}{\log_2 n}+\frac{1}{\log_3 n}+\frac{1}{\log_4 n}+.....+\frac{1}{\log_{100}n}$$

$\log_n 2+\log_n 3+\log_n 4+\ldots.+\log_n 100$

$\log_n 2.3.4\ldots\ldots..100=\log_{100!}100!=1$

Sol.28. (b)

$\log_{10}2\log_2 10+\log_{10}(10^x)=2$

$\frac{\log 2}{\log 10}.\frac{\log 10}{\log 2}+x\log_{10}10=2$

S.
OFFICIUM SUPREMUM

$1 + x = 2$

$\Longrightarrow x = 1$

Sol.29. (c)

$\log_{10} 2$, $\log_{10}(2^x - 1)$, $\log_2(2^x + 3)$ are in AP

$2\log_{10}(2^n - 1) = \log_{10}2 + \log_{10}(2^x + 3)$

$(2^x - 1)^2 = 2.\ (2^x + 3)$

$\Longrightarrow (2^x)^2 - 2.2^x + 1 = 2.2^x + 6$

$\Longrightarrow (2^x)^2 - 4.2^x - 5 = 0$

$\Longrightarrow (2^x - 5)(2^x + 1) = 0$

$\Longrightarrow 2^x = 5,\ 2^x = -1$

If $2^x = 5$

$x \log 2 = \log 5$

$x = \log_2 5$

Sol.30. (c)

$\log_x a$, a^x and $\log_b x$ are in GP

$$(a^x)^2 = \log_x a.\log_b x$$

$$(a^x)^2 = \frac{\log a}{\log x}.\frac{\log x}{\log b}$$

$$(a^x)^2 = \frac{\log a}{\log b} = \log_b a$$

Taking log both side

$\log(a^{2x}) = \log(\log_b a)$

$2x \log a = \log(\log_b a)$

$$2x = \frac{\log(\log_b a)}{\log a} = \log_a(\log_b a)$$

$$x = \frac{\log_a(\log_b a)}{2}$$

Sol.31. (b)

$\log_2(x - 1) = \log_2(x - 3)$

$$\frac{1}{2}\log_2(x-1) = \log_2(x-3)$$

$$\Rightarrow \log_2(x-1) = 2\log_2(x-3)$$

$$\Rightarrow \log_2(x-1) = \log_2(x-3)^2 \quad (x-1) = (x-3)^2$$

$$\Rightarrow x - 1 = x^2 - 6x + 9 \Rightarrow x^2 - 7x + 10 = 0$$

$\Rightarrow (x - 2)(x - 5) = 0$

$x = 2$ or $x = 5$

but $x = 2$ is not possible because domain of logarithmic functions is positive.

so $x = 5$ is only one answer.

Sol.32. (d)

$$\log_x\left(\frac{x}{y}\right) + \log_y\left(\frac{y}{x}\right) = k$$

$$\log_x x - \log_x y + \log_y y - \log_y x = k$$

$$1 - \log_x y + 1 - \log_y x = k$$

$$2 - \log_x y - \log_y x = k$$

$$2 - (\log_x y + \log_y x) = k$$

$$2 - \left(\log_x y + \frac{1}{\log_x y}\right) = k$$

we know that

$$\left(\log_x y + \frac{1}{\log_x y}\right) \geq 2$$

$$-\left(\log_x y + \frac{1}{\log_x y}\right) \leq -2$$

$$2 - \left(\log_x y + \frac{1}{\log_x y}\right) \leq 0$$

so $k \leq 0$

value of k cannot be 1

Sol.33. (a)

$\log_b a = p$

then $a = b^p$ ……….(i)

$\log_d c = 2p$

then $c = d^{(2p)}$ ……….(ii)

$\log_f e = 3p$

then $e = f^{(3p)}$ ……….(iii)

$$(ace)^{\frac{1}{p}} = \left[b^p.d^{2p}f^{3p}\right]^{\frac{1}{p}}$$

$$(ace)^{\frac{1}{p}} = bd^2f^3$$

Sol.34. (c)

$p + q$

$$\sum_{j=1}^{n}\log_{10}2^j + \sum_{j=1}^{n}\log_{10}5^j$$

$$\sum_{j=1}^{n} j\log_{10}2 + \sum_{j=1}^{n} j\log_{10}5$$

$$= \sum_{j=1}^{n} j\log_{10}10 = \sum_{j=1}^{n} j \quad \text{……….(i)}$$

Given that $p + q = 66$

$$\Rightarrow 1 + 2 + 3 + ... + n = 66$$

$$\Rightarrow \frac{n(n+1)}{2} = 66$$

$\Longrightarrow n = 11$

Sol.35. (d)

Given that

$p + q = 15$

now by equation (i)

$$p + q = \sum_{j=1}^{n} j = 15 \quad \text{…………(ii)}$$

$$q - p = \sum_{j=1}^{n}\log_{10}5^j - \sum_{j=1}^{n}\log_{10}2^j$$

$$\Rightarrow \sum_{j=1}^{n} j\log_{10}5 - \sum_{j=1}^{n} j\log_{10}2 = \sum_{j=1}^{n} j\log_{10}\frac{5}{2}$$

$$\Rightarrow \log_{10}\frac{5}{2}\sum_{j=1}^{n} j = \log_{10}2.5 \times \sum_{j=1}^{n} j$$

$$\Rightarrow \log_{10}2.5 \times 15 = 15\log_{10}2.5$$

<u>You can access the solutions of PYQ through 3 Youtube video:</u>

1. logarithm 2011 to 2015 | NDA mathematics previous year questions by Sandeep Brar
2. logarithm 2016 to 2020 | NDA mathematics previous year questions by Sandeep Brar
3. logarithm 2021 to 2025 | NDA mathematics previous year questions by Sandeep Brar

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